

Technical Document #95: Cloud Computing - Comprehensive Overview

Technical Document #95: Cloud Computing - Comprehensive Overview

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Multi-cloud strategies use services from multiple cloud providers.
- Containerization packages applications with their dependencies.
- Serverless computing allows developers to run code without managing servers.